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Genome-wide identification and functional analysis of JAZ gene family in *Rosa hybrida* reveals the positive role of *RhJAZ16* in salt stress tolerance

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ABSTRACT

JASMONATE ZIM-domain (JAZ) proteins serve as central nodes in jasmonic acid pathway to regulate plant growth and environmental responses. Although the JAZ gene family in several crops has been studied, its characterization in *Rosa hybrida* remains limited. In this study, 32 RhJAZ gene members were identified in the *Rosa hybrida*, and their characterization revealed distinct genomic locations, evolutionary relationships, and structural domains. Further cis-elements analysis of promoter showed significant enrichment of jasmonic acid (JA)- and abscisic acid (ABA)-responsive regulatory elements, suggesting potential involvement in hormone crosstalk. Expression profiling across ten tissues and under salt stress conditions revealed diverse expression patterns, with several genes showing tissue specificity and salt-inducible expression. Functional analysis revealed that *RhJAZ16* localizes to the nucleus, is highly expressed in roots, and is significantly induced by salt stress. *RhJAZ16* overexpression in *Arabidopsis thaliana* enhanced salt tolerance by improving seed germination, biomass accumulation, chlorophyll retention, and reducing oxidative damage. This study offers a comprehensive survey of JAZ family members in *Rosa hybrida* and identifies *RhJAZ16* contributing to enhanced salt stress resilience. These findings provide new insights into JA-mediated stress signaling in rose by revealing the enrichment of ABA-responsive cis-elements and the salt-inducible role of *RhJAZ16*, thereby offering a basis for future molecular breeding of stress-resilient ornamental plants.

1. Introduction

Jasmonic acid (JA), a ubiquitous phytohormone, plays essential roles in regulating various developmental processes and environmental responses in plants (Ali and Baek, 2020). Acting downstream of JA signaling, JASMONATE ZIM-domain (JAZ) proteins serve as central repressors modulating nutrient-associated development, cell cycle progression, and adaptive responses to biotic and abiotic stimuli (Pauwels and Goossens, 2011a; Zhao et al., 2024). Given their central role in coordinating developmental and stress responses, exploring the functions of JAZ proteins in ornamental plants such as the modern rose *Rosa hybrida* (*R. hybrida*) is essential, as their roles are well characterized in crops, whereas they remain largely unexplored in roses. These proteins, part of the plant TIFY protein family, attenuate jasmonate signaling outputs by modulating key gene expression cascades (Liu et al., 2021; Thines et al., 2007). Although JAZ proteins display significant sequence divergence, they typically contain three conserved structural elements: an N-terminal region, a ZIM domain, and a Jas motif at C-terminal (Bai et al., 2011; Howe and Yoshida, 2019). The ZIM domain, approximately 30 amino acids in length, is centrally located and includes the highly conserved TIFY sequence, which mediates both repressor activity and protein–protein interactions within the JAZ family (Zhou et al., 2023). In contrast, the Jas motif facilitates binding to the JA co-receptor COI1 and various transcriptional regulators (Katsir et al., 2008).

Although JAZ proteins are widely characterized as repressors in JA signal transduction, their member-specific functions and regulatory networks in plants have yet to be clearly characterized. Recent studies highlight the functional specificity of JAZ proteins in diverse biotic stress responses. For example, JAZ10 is strongly induced upon infection with *Pseudomonas syringae*, and *jaz10–1* mutants exhibit altered disease resistance (Sehr et al., 2010). JAZ proteins also play central roles in plant resistance to herbivorous insects by modulating jasmonate-dependent

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secondary metabolism and defense gene expression (Li et al., 2024a). For instance, in cotton, *GhJAZ24* enhances resistance to *Helicoverpa armigera* and *Spodoptera frugiperda*, and its deployment in an inducible JAZ (iJAZ) system allows pest-specific activation of defense while avoiding growth penalties (Mo et al., 2024; Li et al., 2024a). Beyond biotic stresses, accumulating evidence suggests that several JAZ genes contribute significantly to abiotic stress adaptation, particularly salt stress. Salt stress enhances the biosynthesis of JA, and then activate JA downstream pathway to mediate plant tolerance mechanisms (Zhao et al., 2024). In rice, OsJAZ9 interacts with OsbHLH to regulate downstream salt stress-responding genes (Wu et al., 2015). Moreover, overexpressing *PnJAZ1* in *Pohlia nutans* enhanced the moss's salt stress tolerance via regulation of the abscisic acid (ABA) signaling (Liu et al., 2019). JA signaling promotes COI1-mediated degradation of JAZ8 in *Arabidopsis*, permitting NF-Y complex activity and induction of salt-stress response genes (Li et al., 2024b). Together, these studies illustrate the functional diversity of JAZ proteins and suggest that they are critical players in salt stress responses, thereby underscoring the importance of investigating their genome-wide repertoire and stress-related roles in the commercially important ornamental crop, *R. hybrida*.

Roses (*Rosa* spp.) represent one of the most commercially valuable groups of ornamental horticultural crops worldwide, contributing billions of dollars annually to the global floriculture industry (Darras, 2020). Modern rose (*R. hybrida*), derived from complex hybridization events between Chinese roses and various European species, are distinguished by their continuous flowering, diverse floral traits, and intense fragrance (Bendahmane et al., 2013; Smulders et al., 2019). Driven by these desirable attributes, intensive breeding has resulted in the creation of over 35,000 tetraploid cultivars, and widely grown across the world (Zhang et al., 2024). Notably, rose cut flowers account for over 30 % of the global cut flower market, underscoring their economic significance. Despite this horticultural success, the high degree of heterozygosity and genome complexity in modern rose—resulting from extensive hybridization—poses major challenges for genome assembly and functional genomics. A high-quality haploid genome of 'Samantha' (a tetraploid cultivar) was recently assembled, providing a valuable resource for dissecting gene families and advancing molecular studies in roses (Zhang et al., 2024).

In roses, salt stress markedly suppresses vegetative growth, while also accelerating leaf senescence and chlorosis, thereby reducing the aesthetic quality and commercial value of flowers (Cai et al., 2014; Li et al., 2016). Recent studies have highlighted the pivotal role of JAZ proteins in mediating salinity responses through jasmonate signaling and its crosstalk with other hormones (Zhao et al., 2024). While the JAZ gene family has undergone characterization in numerous economically crops, including maize, tomato, and alfalfa (Han and Luthe, 2021; Sun et al., 2021; Yan et al., 2024), their roles remain unexplored in ornamental plants, particularly roses. This gap is particularly noteworthy for *R. hybrida*, a globally important ornamental species highly sensitive to salinity stress. Therefore, the objective of this study was to systematically identify and characterize the JAZ gene family in *R. hybrida*, analyze their expression under salt stress, and functionally validate representative candidates. This study hypothesize that specific JAZ genes in *R. hybrida* are responsive to salt stress and that their overexpression can confer enhanced tolerance by modulating jasmonate-associated transcriptional networks. To test this hypothesis, we leveraged newly available genomic resources to perform a genome-wide analysis of the JAZ gene family in *R. hybrida*, followed by expression profiling under salt stress and functional validation of *RhJAZ16* in transgenic *Arabidopsis* lines. These findings offer new insights into the function of JAZ genes in *R. hybrida* and establish a foundation for enhancing salt stress tolerance through molecular breeding.

2. Materials and methods

2.1. Identification of JAZ genes in Rosa hybrida

Putative JAZ genes in *R. hybrida* were identified using Hidden Markov Model (HMM) searches against the conserved TIFY (PF06200) and Jas (PF09425) domains retrieved from the Pfam database. These domain profiles were used to scan the 'Samantha' reference genome (Zhang et al., 2024) with HMMER v3.0, employing an E-value threshold of 1e-10. Identified candidates were validated for the presence of complete TIFY and Jas domains using NCBI's CDD tools (Conserved Domain Database) and SMART tools. The isoelectric point (*p*I) and molecular weight of predicted JAZ genes were also estimated via ExPASy ProtParam. Subcellular localization of RhJAZs was inferred using WoLF PSORT tools.

2.2. Phylogenetic and gene structure analysis

Multiple sequence alignments of JAZ proteins from *R. hybrida* and *Arabidopsis thaliana* were conducted by Clustal X (Thines et al., 2007). The neighbor-joining (NJ) method implemented in MEGA5 was applied to generate the phylogenetic tree, with statistical support evaluated by 1000 bootstrap iterations (Tamura et al., 2011). The phylogenetic tree was further visualized by Evolview platform (Subramanian et al., 2019). Gene structures, including exon-intron arrangements, were visualized using the GSDS 2.0 based on genomic sequences (Hu et al., 2015).

2.3. Conserved motif and cis-element analysis

The MEME Suite tool was used to identify conserved motifs with 10 motifs setting and the width of motifs ranging from 6 to 50 amino acids. Promoter regions (2000 bp upstream of the start codon) of RhJAZ genes were scanned for cis-acting element analysis using the PlantCARE. The functional classification of these elements was visualized by TBtools (Chen et al., 2020).

2.4. Synteny analysis

Synteny analysis was conducted using MCScanX to identify segmental and tandem duplications within *R. hybrida*, as well as syntenic relationships with *Arabidopsis thaliana*, *Malus domestica*, and *Prunus persica* (Wang et al., 2012). Chromosomal distribution and intragenomic synteny of RhJAZs were assessed based on *R. hybrida* annotation and visualized by the Circos tool (Krzywinski et al., 2009).

2.5. Growth conditions and stress treatment

Tissue-cultured seedlings of *R. hybrida* cv. 'Samantha' were acclimated for subsequent experimental treatments. After 60 days of in vitro development, seedlings were taken out of culture vessels, and residual growth medium was gently rinsed from the roots using sterile water. The cleaned seedlings were transferred into containers containing a 1:1 (v/v) blend of peat and perlite. Plants were further grown in environment-controlled chambers with a 16 h light/8 h dark photoperiod, a temperature of 23 ± 1°C, a humidity of 70 %, and fluorescent light intensity maintained at 100 μmol m^{-2} s^{-1} .

For the salt treatment, the 60-day-old seedlings were exposed to different concentrations of NaCl (0 mM, 150 mM, 200 mM, and 250 mM). After 20 days of treatment, when the leaves showed symptoms such as edge burning and wilting, the second fully expanded leaf was collected for RNA extraction. For RNA extraction, tissues including stems, leaves, roots, and open flowers were collected from *R. hybrida* cv. 'Samantha', with three independent biological samples obtained per tissue type.

2.6. Transcriptomic analysis of RhJAZs

Publicly available RNA-seq datasets of ten tissues from *R. hybrida* cv. 'Samantha' were downloaded from BioProject PRJNA110816 (Zhang et al., 2024) and used for the tissue-specific expression analysis of *RhJAZ* genes. In addition, transcriptomic data generated under salt stress treatments (0, 100 mM, and 200 mM NaCl) were produced in this study using the DNBSEQ-T7 sequencing platform and deposited in BioProject PRJNA1265865.

All sequencing reads were aligned to the *R. hybrida* cv. 'Samantha' reference genome using HISAT2 (v2.1.0) software with default parameters (Pertea et al., 2016). Gene expression levels were quantified and assembled using StringTie (v1.3.4) (Pertea et al., 2015), and transcript abundance was normalized as FPKM (fragments per kilobase of transcript per million mapped reads). Differential expression of RhJAZ genes across tissue types and salt-treated samples was normalized and graphically represented by TBtools (Chen et al., 2020).

2.7. Generation of transgenic Arabidopsis overexpressing RhJAZ16

To generate *RhJAZ16* overexpressing plants, the complete CDS of *RhJAZ16* was isolated from *R. hybrida* cv. 'Samantha' cDNA and inserted into the pHB expression vector (Liu et al., 2022). The resulting plasmid was delivered into *Arabidopsis thaliana* (Col-0) using the floral dip technique (Clough and Bent, 1998). The seedlings with transformation were selected on hygromycin (50 mg·L^{-1}) and confirmed through RT-PCR (Table S8). Three independent homozygous T_3 lines exhibiting elevated expression were chosen for subsequent analyses.

2.8. Seed germination assay under salt stress

Seeds from *Arabidopsis thaliana* Col-0 and *RhJAZ16*-overexpressing lines were surface sterilized and placed onto 1/2 MS (Murashige and Skoog) agar supplemented with varying NaCl levels (0, 100, 150, and 200 mM). Germination progress was monitored daily for 7 days, with radicle emergence considered as the germination criterion. Each condition included no fewer than three biological replicates, each consisting of 50 seeds per genotype.

2.9. Salt treatment at the seedling stage in Arabidopsis transgenic lines

For salt tolerance assays at the seedling stage, 10-day-old *Arabidopsis* seedlings grown on 1/2 MS medium were transferred to soil and irrigated with either distilled water (control) or 250 mM NaCl solution every 5 days for a period of 15 days. Phenotypic changes were documented through photography, and plants were harvested for physiological measurements. Three biological replicates were used for each measurement, and data are reported as mean ± standard deviation (SD). Statistical significance was assessed using Student's *t*-test.

2.10. Measurement of physiological parameters

The fresh and dry biomass of aerial parts were was recorded following salt treatment. To determine dry weight, samples were oven-dried at 65 °C for 48 h. Chlorophyll content was extracted using 80 % acetone and measured at 645 and 663 nm using a spectrophotometer. Malondialdehyde (MDA) content was determined following the manufacturer's protocol of MDA detection kits (Nanjing Jiancheng, China). Three biological replicates were used for each measurement, and data are reported as mean ± SD. Statistical significance was assessed using Student's *t*-test.

2.11. RNA isolation and quantitative PCR analysis

Total RNA was isolated from plant tissues using the kit from Tiangen Co., Ltd. (China), following the provided protocol. Complementary DNA (cDNA) synthesis was performed using the kit from TaKaRa Co., Ltd. (China), following the provided protocol. Quantitative real-time PCR (qRT-PCR) was subsequently performed and all reactions were run in triplicate using three separate biological samples to ensure data consistency. *RhUBI2* (*Ubiquitin-2*) was used as the internal control and expression levels were quantified using the $2^{-\Delta\Delta CT}$ method (Livak and Schmittgen, 2001). The stability of *RhUBI2* under our experimental conditions was evaluated using the RefFinder program (Xie et al., 2023), confirming its suitability for normalization. Primers were designed using the Primer-BLAST online tool and are listed in Table S8.

2.12. Subcellular localization

The CDSs of *RhJAZ16* were cloned into the pHB-GFP vector to construct 35S:*RhJAZ*-GFP. The 35S:*RhJAZ*-GFP and the empty vector pHB-GFP (control) were transformed into mesophyll protoplasts of *Arabidopsis thaliana* using PEG (polyethylene glycol)-mediated methods (Yoo et al., 2007). GFP fluorescence was detected by confocal laser scanning microscope (Zeiss LSM 710). The excitation and emission wavelengths for GFP were set to 488 nm and 510–530 nm, respectively.

3. Results

3.1. Identification of the JAZ genes in Rosa hybrida

To identify the JAZ gene family in modern rose (*R. hybrida)*, BLAST alignment and domain screening methods were employed. After removing sequences that lacked both the TIFY and JAS domains, as well as redundant fragments, 32 JAZ genes were identified from the haplotype-resolved assembly of the tetraploid 'Samantha' genome, in which all four haplotypes are separately represented (Table S1). These genes were designated as *RhJAZ1* to *RhJAZ32*, based on their physical order along the haplotype-resolved pseudo-chromosomes (1A–7D), from top to bottom. Among the 32 genes, the sizes of the RhJAZ proteins display considerable variation in sequence length, spanning from 132 to 390 amino acids, and exhibit predicted molecular masses ranging from 14.88 to 41.61 kDa. The RhJAZ genes are distributed across four homologous chromosome groups of modern rose, and their predicted isoelectric points (*p*I) values range from 7.62 to 10.21, suggesting that they are predominantly basic and potentially enriched in positively charged residues, such as lysine and arginine. According to in silico predictions, the majority of RhJAZ proteins (62.5 %) are likely localized in the nucleus, whereas the remaining 37.5 % are predicted to reside in the chloroplast.

3.2. Phylogenetic relationships and classification of JAZ proteins

To investigate the potential functional diversification of JAZ proteins, we constructed a phylogenetic tree of JAZ family members from five representative plant species: *R. hybrida* (Rh), *Arabidopsis thaliana* (At), *Oryza sativa* (Os), *M. domestica* (Md), and *P. persica* (Prupe) (Table S2). The phylogenetic tree with robust clustering supports the evolutionary grouping of these genes. A total of 84 JAZ proteins were clearly classified into five distinct subfamilies (Groups I–V), highlighting the conserved and diversified evolutionary relationships of the JAZ family across plant lineages (Fig. 1). Group IV and V were the largest and most broadly represented, including JAZ proteins from all five species. This widespread conservation implies that genes in these Groups are likely involved in core jasmonate signaling functions that are maintained across angiosperms. Group III also displayed relatively high interspecies conservation, particularly among members of the Rosaceae family (*Malus*, *Prunus*, *Rosa*), suggesting potential functional orthology related to growth and developmental regulation. In contrast, Group I exhibited signs of lineage-specific expansion, particularly in *R. hybrida*, which contributed multiple closely related paralogs. This pattern may reflect recent duplication events followed by subfunctionalization or

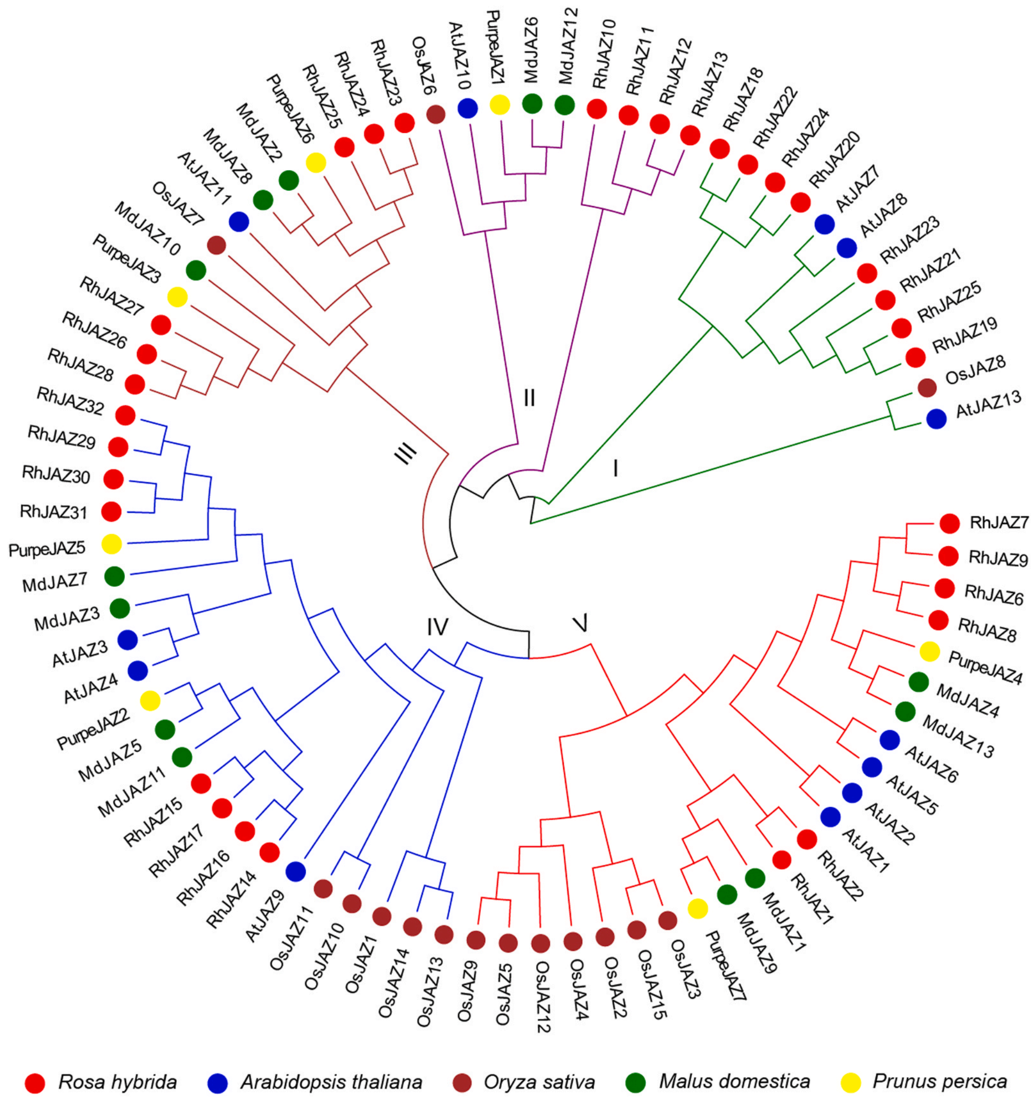

Fig. 1. Phylogenetic relationship analysis of JAZ family. Phylogenetic relationships among JAZ proteins from five plant species—including *R. hybrida* (Rh, red), *Arabidopsis thaliana* (At, blue), *Oryza sativa* (Os, brown), *Malus domestica* (Md, green), and *Prunus persica* (Prupe, yellow). Five groups were divided from the phylogenetic tree, with each clade represented by a different color.

neofunctionalization in response to environmental or developmental cues unique to this species. The phylogenetic structure aligned well with known taxonomic relationships. Members from Rosaceae (*R. hybrida*, *M. domestica*, *P. persica*) frequently clustered together within the same Groups, suggesting common ancestral origins and conserved roles in woody perennial development.

3.3. Phylogeny, gene structure, and conserved motif analysis

Structural diversity and conserved elements within RhJAZs were investigated through detailed analyses of exon–intron organization and conserved motif composition (Fig. 2A). Based on phylogenetic analysis, the 32 identified RhJAZ genes were organized into 9 non-redundant homeologous groups, each representing copies from the four haplotypes of the tetraploid genome. These groups were further classified into three major clades (Clades I–III), highlighting lineage-specific diversification within *R. hybrida*. Exon–intron structure analysis revealed notable variation among RhJAZ genes (Fig. 2A, center panel). Most *RhJAZ* genes contained between two and six exons, but considerable structural diversity was observed across the three clades. Members of Clades I and III generally possessed a greater number of exons, suggesting a more complex gene organization. For example, *RhJAZ15* and *RhJAZ29* (Clade I) contained 6–7 exons, while *RhJAZ13* and *RhJAZ12* (Clade III) harbored markedly long intronic regions, which may be associated with transcriptional regulation or alternative splicing events, contributing to potential functional versatility. In contrast, Clade II members (e.g., *RhJAZ18*, *RhJAZ24*) typically harbored only one or two exons, indicating a more compact genomic structure.

To further explore the conserved features and potential functional divergence of RhJAZ, MEME analysis revealed ten distinct conserved motifs among the RhJAZ proteins (Fig. 2B and Fig. S1). While all RhJAZ proteins shared Motif 1 and Motif 3—corresponding to the conserved Jas and ZIM domains—each clade exhibited distinct motif compositions. These differences suggest that clade-specific motif combinations may contribute to functional diversification within the RhJAZ family. Clades I and III displayed a broader and more complex motif architecture, potentially indicating more sophisticated regulatory or interaction capabilities. In contrast, Clade II proteins exhibited a reduced number of motifs and uniquely contained Motif 9, which may confer specific functional adaptations. The observed variations in exon–intron structures and motif architectures reflect the evolutionary flexibility of RhJAZ genes.

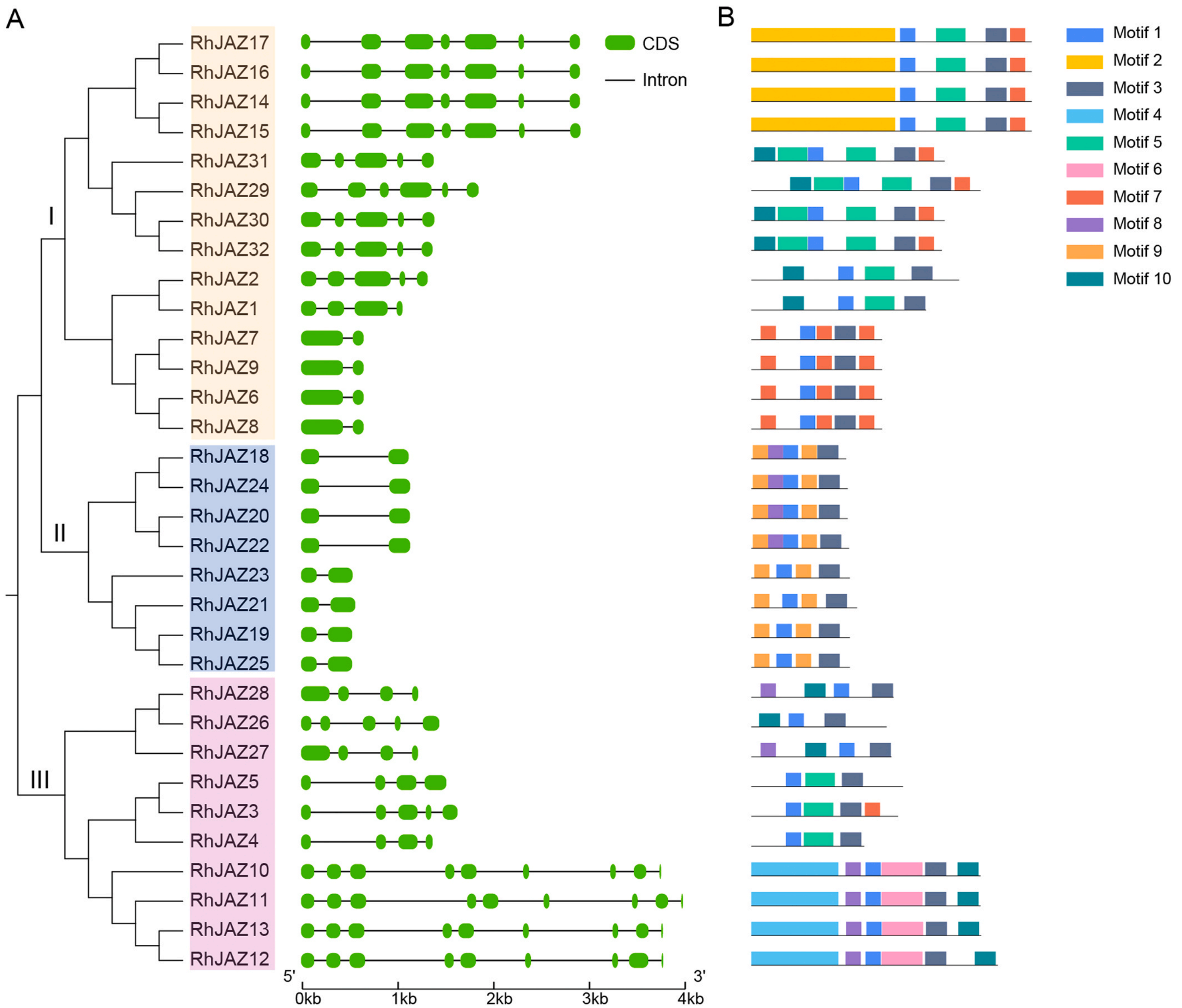

Fig. 2. Phylogenetic relationships, gene structures, and conserved motifs of RhJAZ genes in *R. hybrida*. (A) Phylogenetic tree and exon–intron structures of RhJAZ genes. The neighbor-joining (NJ) tree on the left illustrates the evolutionary relationships among 32 RhJAZ proteins, which cluster into three major clades (I–III) based on their complete amino acid sequences. The middle panel displays the exon–intron organization of each RhJAZ gene, with green boxes representing exon and black lines indicating introns. Gene structures are aligned according to the phylogenetic tree. (B) Distribution of conserved motifs in 32 RhJAZ proteins. A total of 10 conserved motifs is shown as colored boxes. Each motif is indicated by a different color and numbered on the right. The length and arrangement of motifs reflect the diversity and conservation among different RhJAZ protein members.

3.4. Cis-Element Composition and Functional Categorization of RhJAZ Promoters

The 2000 bp upstream sequences of each RhJAZ gene were examined for cis-regulatory elements to explore their potential mechanisms of transcriptional regulatory (Table S3). The results revealed a wide distribution of cis-elements related to hormone responses, stress-related adaptation, and plant developmental regulation across the RhJAZ promoters (Fig. 3A). To further interpret their potential functional roles, these elements were categorized into five major groups: hormone-, stress-, light-, and development- responsive elements (Fig. 3B).

All 32 RhJAZ genes harbored a high number of light-responsive elements, indicating that the RhJAZ family may play important roles in light-mediated signaling pathways in modern rose. Most genes also contained multiple cis-elements responsive to MeJA (methyl jasmonate), auxin, ABA, and SA (salicylic acid), suggesting extensive involvement of RhJAZ genes in diverse hormone signaling pathways. Notably, MeJA-responsive elements were enriched across all three phylogenetic clades, in line with the established role of JAZ proteins as key regulators in the JA signaling pathway. Among hormone-related elements, ABA-responsive motifs were the most abundant, with Clade I and Clade II genes showing significantly higher average numbers (8.8 per gene) compared with Clade III (2.3 per gene). This pattern suggests that the Clades I and II genes likely more prominently involved in ABA-mediated regulatory processes. Additionally, stress-associated cis-elements were prevalent, although their distribution varied markedly among genes. Low-temperature response elements—including low-temperature responsive motifs and HvMYB1-binding sites—were the most common, present in 26 RhJAZ genes. For example, *RhJAZ10*, *RhJAZ11*, and *RhJAZ12* contained relatively higher numbers of cold-responsive elements, implying a potential role in cold stress adaptation. In addition, a subset of genes, including *RhJAZ3–RhJAZ5*, *RhJAZ14–RhJAZ17*, and *RhJAZ29–RhJAZ32*, possessed development-related cis-elements, such as those associated with meristem-specific

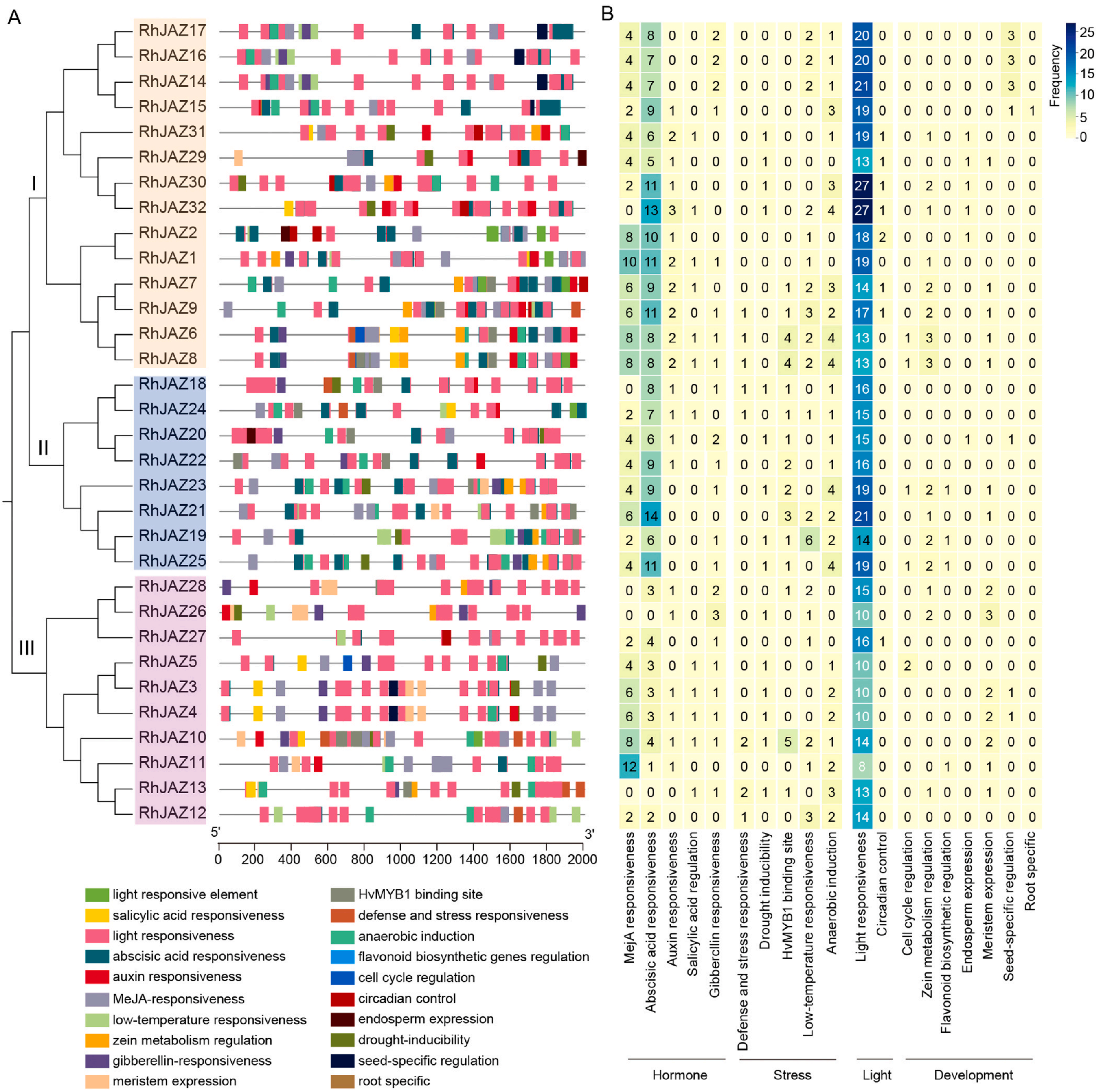

Fig. 3. Cis-acting regulatory elements in the promoter regions of RhJAZ genes. (A) Distribution of predicted cis-elements of RhJAZ genes. Colored boxes represent different types of cis-elements. Gene names are arranged according to the phylogenetic tree on the left. (B) Heatmap showing the number of each cis-element type identified in RhJAZs promoters, grouped into functional categories including hormone, stress, light, and development. The color gradient represents element abundance, with darker colors indicating higher frequencies.

expression, endosperm expression, and seed-specific expression. These cis-elements indicate potential regulatory roles of the corresponding genes in floral development and reproductive function, possibly exhibiting clade-specific patterns of expression.

3.5. Synteny analysis of RhJAZ genes

To investigate gene duplication events within the RhJAZ family, we then performed a collinearity analysis of the 32 RhJAZ genes. Gene duplication can occur via segmental duplication, involving interchromosomal events, or tandem duplication, involving adjacent genes on the same chromosome. A total of 28 segmentally duplicated gene pairs were identified, whereas no tandem duplications were detected (Fig. 4A and Table S4). RhJAZ genes exhibited an uneven chromosomal distribution across the *R. hybrida* genome. Chromosomes 2B, 5B, 5D, and 7 C harbored a higher density of RhJAZ genes, consistent with regions of segmental duplication. These results suggest that the expansion of the RhJAZ gene family was primarily driven by large-scale segmental duplication, rather than localized tandem duplication, potentially through genome rearrangement and interchromosomal events.

We further evaluated the evolutionary conservation of RhJAZ genes through synteny analysis with three representative plant genomes (Fig. 4B and Table S5). The analysis identified 32 gene pairs between *Rosa* and *Arabidopsis*, with strong syntenic signals primarily located on

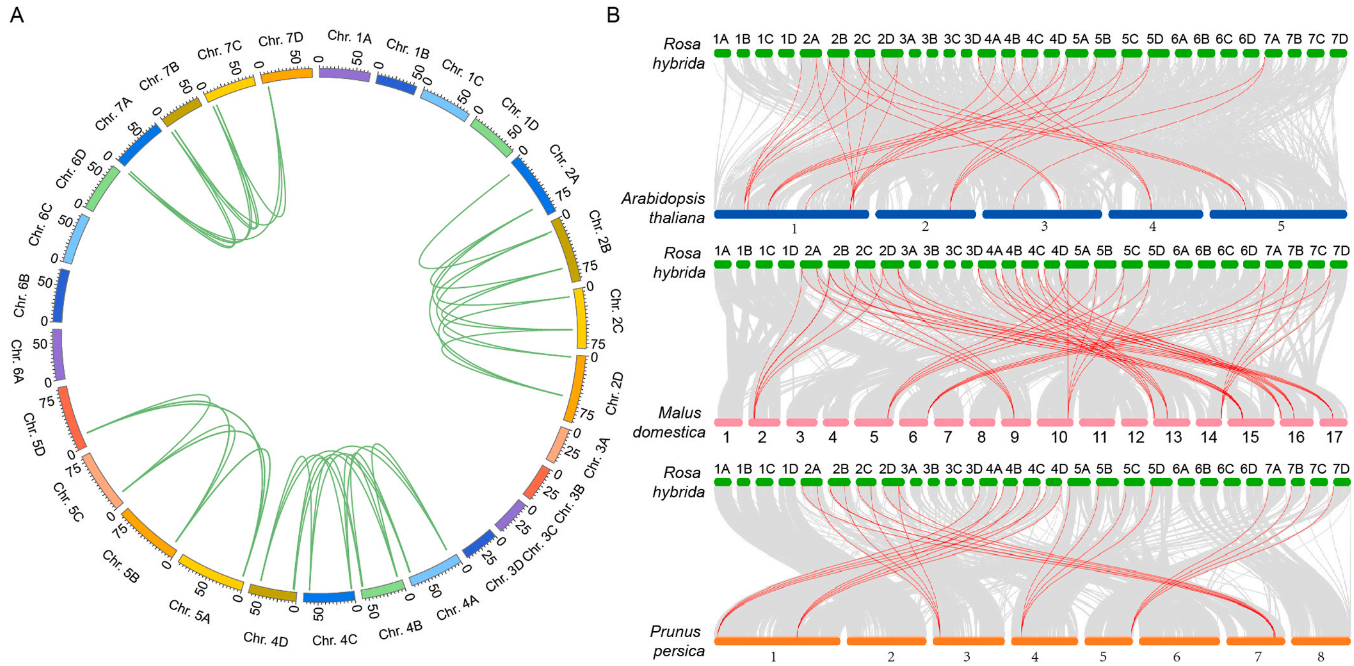

Fig. 4. Chromosomal distribution and synteny analysis of RhJAZ genes. (A) Chromosomal locations of RhJAZ genes on the 28 pseudo-chromosomes. Each chromosome is represented by a colored arc, and green lines indicate segmental duplications among RhJAZs. (B) Synteny analysis of RhJAZ genes between *R. hybrida* and three other species: *Arabidopsis thaliana*, *Malus domestica*, and *Prunus persica*. Green boxes represent RhJAZ genes on modern rose chromosomes, and lines connecting to other species (blue for *Arabidopsis*, pink for *Malus*, orange for *Prunus*) indicate orthologous gene pairs. Red curves highlight conserved syntenic relationships, while gray curves represent background synteny.

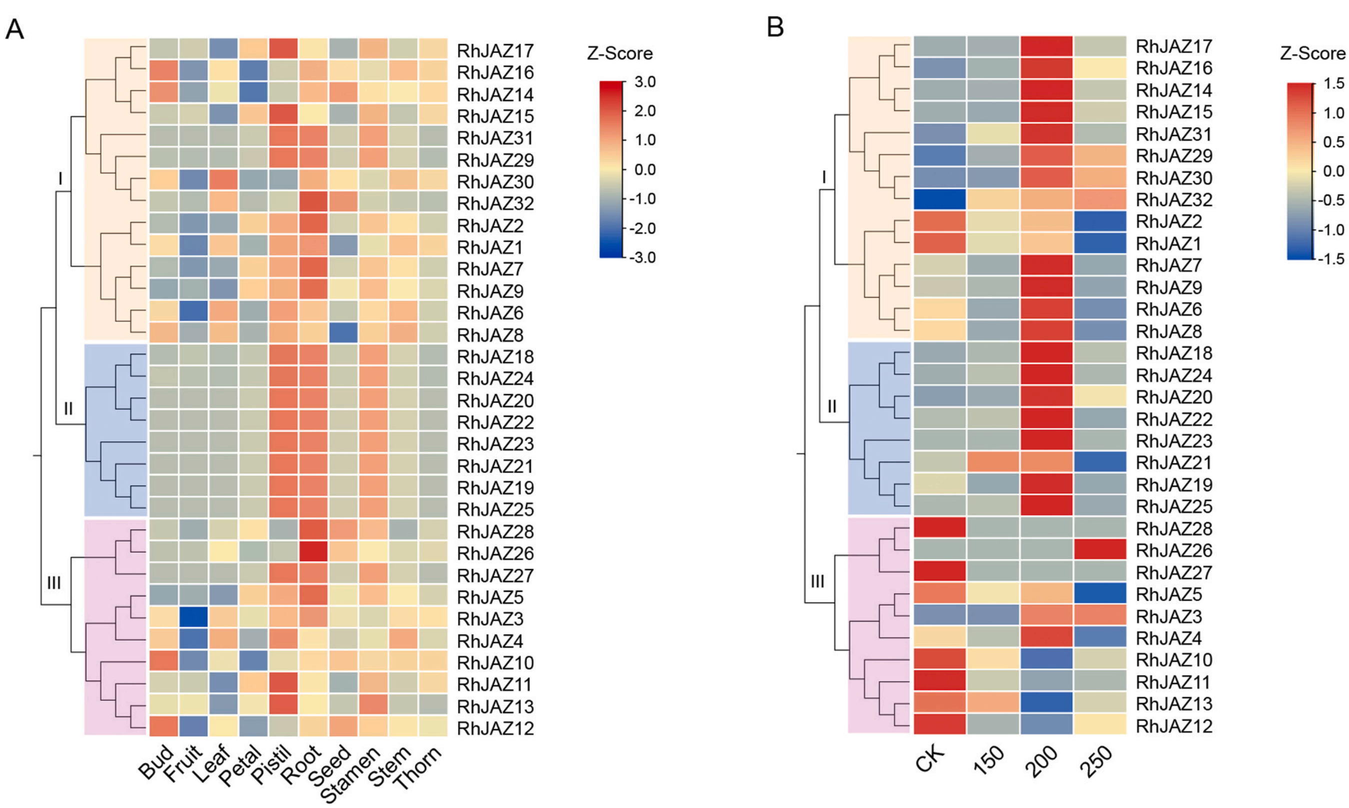

Fig. 5. Expression profiling of RhJAZ genes in *Rosa hybrida* under different tissues and salt stress. (A) Heatmap showing the expression levels of RhJAZ genes across 10 different tissues or organs, including bud, fruit, leaf, petal, pistil, root, seed, stamen, stem, and thorn. Gene expression values were normalized and shown as Z-scores. (B) Heatmap of RhJAZ gene expression under different NaCl treatments. CK represents the control (no salt treatment), and 150, 200, and 250 correspond to treatments with 150 mM, 200 mM, and 250 mM NaCl, respectively. Gene expression values were normalized and shown as row-scaled Z-scores.

Rosa chromosomes 2 and 5. This indicates that JAZ genes are highly conserved among dicotyledonous plants. The syntenic relationship between *Rosa* and *Malus* was even more pronounced, with 51 orthologous gene pairs detected. Tight collinearity was observed between *Rosa* chromosomes 5D, 6B, and 7B and several *Malus* chromosomes, reflecting the close phylogenetic relationship between the two Rosaceae species. Similarly, comparison with *Prunus persica* revealed 26 orthologous gene pairs, with strong syntenic blocks on chromosomes 2 A, 5D, and 6B of *R. hybrida*. These findings further support the notion that *JAZ* genes are evolutionarily conserved across Rosaceae species, while also exhibiting lineage-specific divergence.

3.6. Tissue-specific and salt-induced expression patterns of RhJAZs

Expression profiling of all 32 RhJAZ genes was conducted in ten tissue types of *R. hybrida*, including reproductive (floral buds, pistils, stamens, petals, fruits, and seeds) and vegetative (leaves, roots, stems, and thorns) organs, to infer their potential biological roles (Fig. 5A and Table S6). The resulting heatmap reveals distinct expression patterns, with certain genes showing elevated expression in specific tissues. Most notably, the heatmap indicates that pistil, root, and stamen tissues exhibit the highest expression levels of RhJAZ genes. Clade I and Clade III genes, such as *RhJAZ10*, *RhJAZ12*, *RhJAZ17*, and *RhJAZ16*, exhibit more variable expression across tissues, with particularly prominent expression in reproductive tissues, including buds, petals, and pistils. In contrast, Clade II genes display low basal expression across most tissues, with higher expression levels observed only in pistil, root, and stamen, indicating that these genes may be more specialized and potentially involved in specific developmental stages related to these tissues.

We further analyzed their transcript levels using RNA-seq data from the leaves of 'Samantha' after 15 days of continuous NaCl treatment at 150, 200, and 250 mM (Fig. 5B and Table S7). Clade II genes, along with several Clade I members, showed clear and gradual upregulation at 150 and 200 mM NaCl, with peak expression typically occurring at 200 mM. This pattern indicates their active involvement in salt stress response. However, expression levels of these genes generally declined under 250 mM NaCl, suggesting potential feedback inhibition under excessive salt stress. In contrast, Clade III genes were predominantly downregulated across all salt treatments. Only *RhJAZ3*, *RhJAZ4*, and *RhJAZ26* exhibited slight induction, whereas the majority of Clade III members

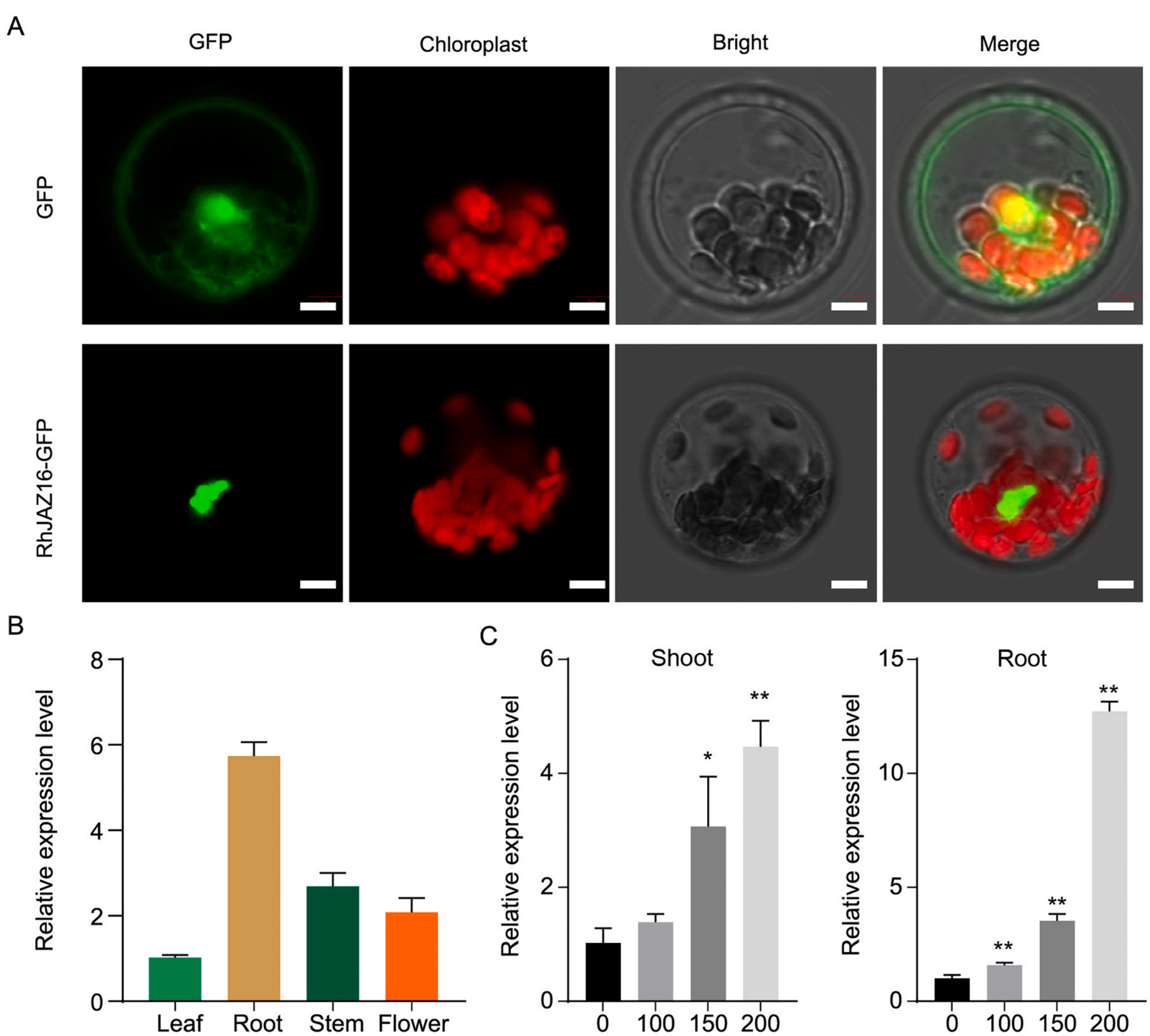

Fig. 6. Subcellular localization and expression analysis of *RhJAZ16* in *R. hybrida*. (A) GFP-tagged RhJAZ16 was transiently expressed in Arabidopsis mesophyll protoplasts to assess subcellular localization. Fluorescent signals for GFP (green) and chlorophyll autofluorescence (red) were captured, along with corresponding bright-field and merged images. The RhJAZ16-GFP signal was predominantly localized in the nucleus. Scale bars = 10 μm. (B) Tissue-specific expression pattern of *RhJAZ16* in *R. hybrida* determined by qRT–PCR. *RhJAZ16* exhibited the highest expression level in roots, followed by stem, flower, and leaf tissues. (C) Expression patterns of *RhJAZ16* in shoots and roots under salt stress. The seedlings of cv. 'Samantha' were treated with 0, 100, 150, and 200 mM NaCl. Transcript levels were measured by qRT–PCR and normalized to an internal control *RhUBI2* gene. Data are presented as means ± SD from three biological replicates. Asterisks indicate statistically significant differences compared with the untreated control: $^{*}P < 0.05$, $^{**}P < 0.01$ (Student's *t*-test).

showed progressive repression. This tissue-specific expression pattern suggests that the RhJAZ family may play a significant role in reproductive development and root function, a hypothesis supported by the presence of development-related cis-elements in their promoters (Fig. 3).

3.7. Expression and functional characterization of RhJAZ16

To further explore the role of RhJAZ genes in salinity adaptation, we selected RhJAZ16 for functional validation. Phylogenetically, its clusters within Group IV alongside functionally characterized orthologs from apple *MdJAZ2* (corresponding to *MdJAZ11* in this study) and rice *OsJAZ10*, both of which positively regulate abiotic stress tolerance (An et al., 2017; Wang et al., 2025). In addition, RhJAZ16 shows relatively high expression in roots and leaves with promoters enriched in JA- and ABA-responsive cis-elements, and it is strongly induced by 200 mM NaCl in transcriptome analyses. Firstly, we detected the subcellular localization of RhJAZ16 in *Arabidopsis* protoplasts. The GFP signal was predominantly localized to the nucleus (Fig. 6A), indicating that RhJAZ16 may function in nuclear signaling processes, likely as a transcriptional regulator involved in stress-responsive gene expression.

Furthermore, qRT-PCR analysis was performed across several plant tissues, including root, leaf, stem, and flower. *RhJAZ16* was detected across all tested tissues, exhibiting peak expression in roots, followed by stems and flowers, while the lowest transcript levels were observed in leaves (Fig. 6B). These findings suggest that *RhJAZ16* is predominantly

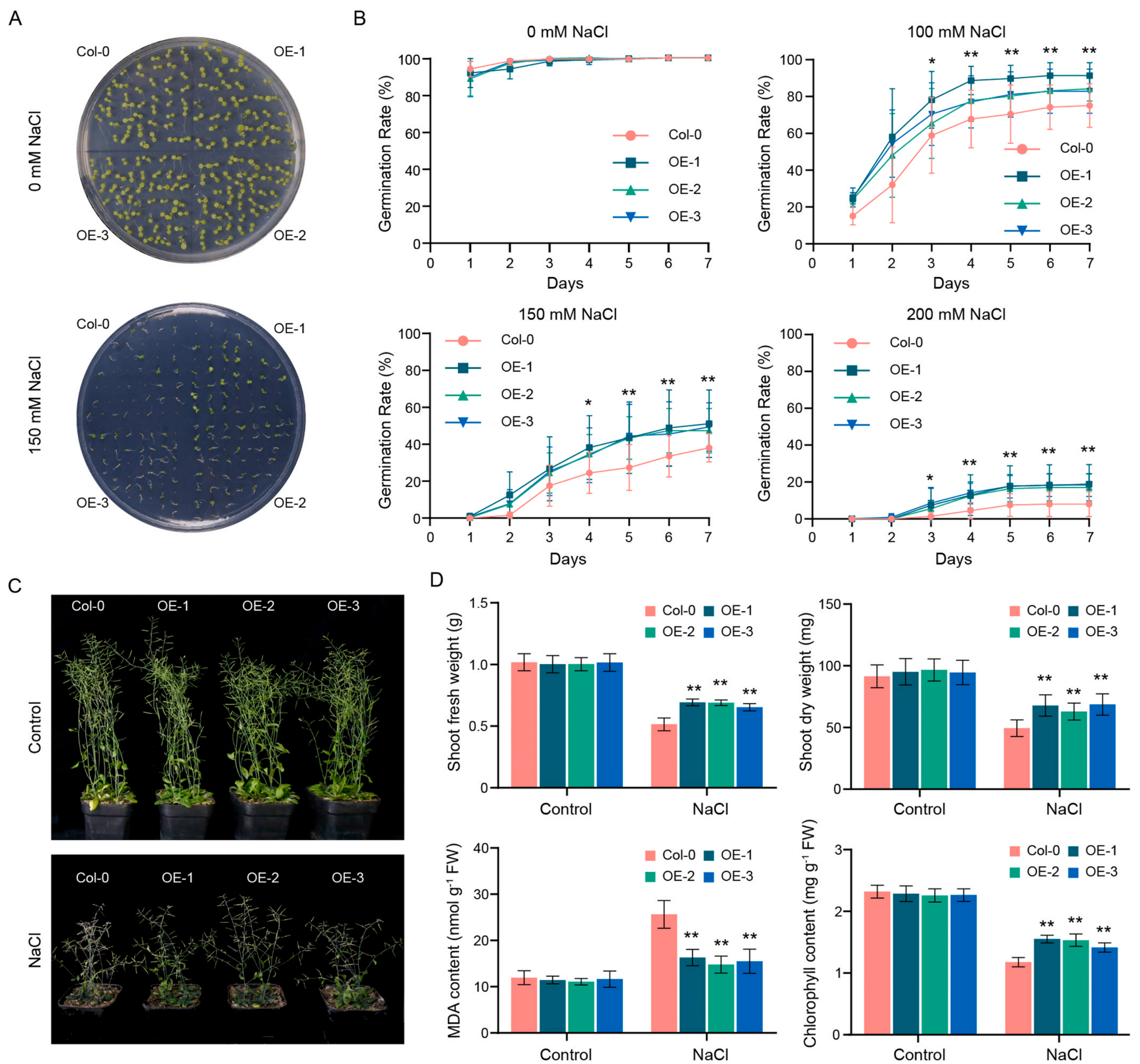

Fig. 7. Overexpression of *RhJAZ16* enhances salt tolerance in *Arabidopsis*. (A) Seed germination of wild-type (Col-0) and three independent *RhJAZ16*-overexpressing lines (OE-1, OE-2, OE-3) under 0 mM and 150 mM NaCl conditions after 7 days. (B) Germination rate of each genotype under 0, 100, 150, and 200 mM NaCl over a 7-day period. Data represent mean ± SD (n = 6). Asterisks indicate statistically significant differences from Col-0 (*$P < 0.05$, **$P < 0.01$).
(C) Growth phenotypes of Col-0 and OE lines at the seedling stage under control and 250 mM NaCl treatment for 15 days. (D) Physiological indices of seedlings under salt stress: shoot fresh weight, shoot dry weight, chlorophyll content, and MDA content. Data represent means ± SD (n = 3). Asterisks denote significant differences compared with Col-0 under the same treatment (**$P < 0.01$).

involved in root-specific processes and may play a key role in root development or in mediating root-specific responses to stress. To assess salt responsiveness, *RhJAZ16* transcript abundance was examined in shoot and root tissues subjected to escalating NaCl concentrations (0, 100, 150, and 200 mM). *RhJAZ16* expression showed a dose-dependent increase in both tissues. In the shoot, expression was significantly induced at 150 and 200 mM NaCl, while in the root, the increase was more pronounced, especially at higher concentrations (Fig. 6C). These results demonstrate that *RhJAZ16* is highly responsive to salt stress, particularly in the root, and suggest its involvement in mediating adaptive responses to saline conditions.

3.8. Overexpression of RhJAZ16 improved salt stress tolerance in Arabidopsis

Three independent *RhJAZ16*-overexpression lines driven by the CaMV 35S promoter (OE-1, OE-2, and OE-3) were confirmed and selected for phenotypic assays (Fig. S2). Germination and early growth performance were assessed under salt stress treatment using NaCl-supplemented MS media, with both wild-type (WT) and the OE lines seeds tested in parallel (Fig. 7A, B). The germination rates of all genotypes were nearly 100 %, with no significant differences observed between Col-0 and the OE lines under 0 mM NaCl (normal conditions). However, under salt stress conditions (100, 150, and 200 mM NaCl), the OE-1, OE-2, and OE-3 lines exhibited higher germination rates than that of Col-0 throughout the 7-day period. In particular, under 200 mM NaCl, the germination rates of the three *RhJAZ16*-OE lines (55 %, 64 %, and 44 %) were significantly higher compared with WT (13 %). These results suggest that *RhJAZ16* overexpression enhances germination capacity under saline conditions.

At the seedling stage, transgenic lines and Col-0 plants displayed similar growth under control conditions. However, after 15 days of salt stress treatment, the WT plants exhibited more severe growth inhibition and chlorosis, while the *RhJAZ16*-OX plants maintained better growth and greener foliage (Fig. 7C). Physiological measurements further confirmed the enhanced stress tolerance of the *RhJAZ16*-OX lines (Fig. 7D). The *RhJAZ16*-OE lines displayed significantly increased fresh and dry shoot biomass, along with higher chlorophyll accumulation under salt stress, while MDA content was significantly lower compared with Col-0 (Fig. 7D). These findings indicate that *RhJAZ16* overexpression enhances salt stress tolerance by improving seed vigor, maintaining photosynthetic pigment stability, and reducing oxidative damage.

4. Discussion

As pivotal components of the JA signaling pathway, JAZ proteins function as transcriptional repressors that regulate various physiological processes (Zhao et al., 2024). In *R. hybrida*, we identified 32 JAZ genes from a haplotype-resolved tetraploid genome, in which all four haplotypes are separately retained. Phylogenetic and synteny analyses grouped these into 9 non-redundant homeologous groups—fewer than the 15–18 members typically found in many diploid species (Du et al., 2024; Han and Luthe, 2021; Pauwels and Goossens, 2011; Shrestha and Huang, 2022). This reduction likely reflects post-polyploid gene loss and fractionation following whole-genome duplication (Domazet-Lošo et al., 2024). Consistently, other species also exhibit relatively small JAZ families irrespective of genome size or ploidy, such as grape (11 members) and watermelon (8 members) (Zhang et al., 2012; Yang et al., 2019). These results indicate that JAZ family size does not scale directly with ploidy but is shaped by evolutionary constraints and functional requirements (Zhao et al., 2024). The classification of five groups in 32 RhJAZ genes consistent with the classification in other dicots (Liu et al., 2021). Notably, Group I contained members exclusively from *Arabidopsis*, rice, and *Rosa*, whereas other Rosaceae species lacked representatives in this lineage, suggesting lineage-specific retention or loss. The tissue-specific enrichment of *R. hybrida* Group I genes in floral organs (Fig. 5A) raises the possibility that these genes contribute to species-specific regulation of jasmonate-mediated floral development, reflecting adaptive specialization within rose. Gene structure and conserved motif analyses also indicated subgroup diversification, though the canonical TIFY and JAS domains remain highly conserved, underscoring their indispensable role in JA signaling (Pauwels and Goossens, 2011; Yan et al., 2024). Synteny analysis revealed the segmental duplication events have driven the expansion of RhJAZ gene family predominantly, rather than tandem duplications. This pattern mirrors observations in alfalfa (Yan et al., 2024), suggesting that genome rearrangements and polyploidization have played key roles in driving JAZ gene family expansion (Zhang et al., 2023). Comparative analyses with *M. domestica* and *P. persica* revealed extensive collinearity and numerous orthologous gene pairs, underscoring the evolutionary conservation of JAZ genes within Rosaceae. The stronger syntenic similarity with apple and peach supports their closer phylogenetic affinity with rose and aligns with recent comparative genomic studies highlighting both conserved regulatory modules and lineage-specific diversification in Rosaceae genomes (Potter et al., 2007; Soundararajan et al., 2019). These results suggest that the current JAZ repertoire in rose likely reflects an evolutionarily conserved set of duplications within Rosaceae, highlighting their potential importance in stress adaptation and developmental regulation.

Jasmonates are vital regulators of plant defense and adaptation, orchestrating responses to diverse biotic and abiotic stresses (Fahad et al., 2015b). Exogenous application of JA has been shown to improve crop performance under adverse conditions such as salinity and drought (Ahmad et al., 2016). For instance, JA treatments enhanced cotton productivity under water deficit stress (Nazim et al., 2021). Consistent with this, JAZ proteins—acting as key repressors downstream of JA signaling—harbor JA-responsive cis-elements in their promoters in our study, suggesting that their transcriptional regulation is tightly coupled with JA-mediated stress responses. Notably, our cis-element analysis revealed that RhJAZ promoters are also enriched in ABA-responsive motifs. Given the central role of ABA in seed germination and abiotic stress adaptation (Nakashima and Yamaguchi-Shinozaki, 2013), this dual enrichment strongly suggests that RhJAZs may integrate JA and ABA pathways to fine-tune stress responses (Fahad et al., 2015b). Mechanistic studies in rice showed that OsJAZ10 interacts with OsMYC2 to repress transcription of ABA/JA-responsive genes such as OsNCED2 and OsRAB21 (Wu et al., 2024), while in *Arabidopsis*, JAZ8 directly represses NF-Y complexes but is degraded under salt-induced JA signaling, thereby releasing ABA/JA-responsive gene expression and enhancing salt tolerance (Zhang et al., 2024). Together, these findings suggest that JAZ proteins act as integrators of ABA and JA signaling under salt stress. This provides a mechanistic framework supporting our observation that *RhJAZ16* overexpression enhances salt tolerance in *Arabidopsis*, and raises the possibility that *RhJAZ16* may contribute to salt tolerance through similar cross-talk mechanisms in rose.

Tissue-specific expression analysis revealed that RhJAZ genes display marked spatiotemporal variation, with particularly high transcript levels in pistils, roots, and stamens. This suggests key regulatory roles in both reproductive development and root-associated functions. Supporting this notion, recent studies demonstrated that JAZ proteins modulate jasmonate levels during petal senescence (Serrano-Bueno et al., 2022), and that RhCIPK3-mediated phosphorylation of RhJAZ5 controls its stability to regulate flower senescence (Chen et al., 2023). Under salt stress, several Clade I/II RhJAZ genes exhibited peak-type induction at 200 mM NaCl with attenuated responses at 150 or 250 mM. Similar threshold-dependent JAZ activation has also been observed in sorghum and alfalfa, where specific members show peak-like induction at particular time points under salt stress rather than gradual changes over time (Du et al., 2022; Li et al., 2025). This likely reflects an optimal window in which JA signaling is most productively engaged while feedback and cross-pathway interactions (e.g., with ABA) restrain responses under lower or higher stress intensities (Fahad et al., 2015b; Ghorbel et al., 2021). Importantly, *RhJAZ16* typifies the inducible Clade I/II response—its root-preferential expression is strongly elevated at 200 mM NaCl in both RNA-seq and qRT-PCR assays, and its overexpression in *Arabidopsis* confers markedly enhanced salt tolerance.

Our functional validation of *RhJAZ16* highlights its pivotal role in salinity tolerance. Transgenic *Arabidopsis* plants overexpressing *RhJAZ16* exhibited markedly improved growth and physiological performance under salt stress, including elevated chlorophyll content and reduced MDA levels compared with wild-type controls. These phenotypic improvements point to an enhanced antioxidative capacity, a mechanism closely tied to hormonal regulation. Indeed, JAs are well known to strengthen reactive oxygen species (ROS)-scavenging systems, with exogenous JA treatments intensifying the activities of superoxide dismutase (SOD), catalase (CAT), and ascorbate peroxidase (APX), while combined JA and ABA signaling alleviates oxidative damage and lipid peroxidation under drought and salinity stress in several species, including *Arabidopsis*, wheat, and pearl millet (Awan et al., 2021; de Ollas et al., 2015; Siddiqi and Husen, 2019). Furthermore, our data suggest that *RhJAZ16* plays a positive regulatory role under salinity stress, although JAZ proteins are generally considered repressors of JA signaling. This functional divergence is consistent with earlier reports: ectopic expression of GhJAZ2 in cotton increased salt sensitivity by repressing JA–MYC2 signaling (Sun et al., 2017), whereas PnJAZ1 in moss enhanced salt tolerance through ABA-regulated pathways (Liu et al., 2019). The enrichment of ABA-responsive cis-elements in the *RhJAZ16* promoter further supports its potential role at the intersection of JA and ABA signaling. Future studies should identify the upstream regulators and downstream targets of RhJAZ16 and clarify its role in JA–ABA crosstalk by validating its function in rose through transgenic or CRISPR-based approaches.

Our findings not only clarify the mechanistic basis of *RhJAZ16*-mediated salt tolerance but also provide direct implications for rose improvement. Salinity remains a major constraint in rose cultivation, particularly under greenhouse and field conditions with saline irrigation. Because *RhJAZ16* enhances root-specific stress adaptation while being linked to reproductive development, it represents a promising target for molecular breeding and genome editing to develop salt-tolerant *R. hybrida* cultivars without compromising floral traits. Furthermore, given that phytohormones interact extensively with the root-associated microbiome and nutrient elements to shape stress adaptation (Fahad et al., 2015a; Singhal et al., 2023), it will be valuable to explore whether *RhJAZ16*-mediated JA signaling intersects with external cues. For instance, stress-induced changes in root metabolites can restructure microbial communities and enhance resilience (Yuan et al., 2024). Taken together, our findings suggest that improving salt tolerance in rose will benefit most from integrated strategies, in which *RhJAZ16* can serve as a candidate gene, complemented by hormonal, microbial, and nutritional interventions.

5. Conclusion

In summary, this study systematically characterized the genomic features of 32 *RhJAZ* genes in the *R. hybrida* genome, including their exon–intron architecture, conserved sequence motifs, and evolutionary relationships. Expression profiling across diverse tissues and salt stress conditions revealed distinct expression patterns, suggesting their functions in developmental regulation and environmental responsiveness. Functional analysis of *RhJAZ16* demonstrated that its overexpression in *Arabidopsis thaliana* confers enhanced salt stress tolerance, likely through stabilization of the photosynthetic apparatus and attenuation of oxidative damage. These findings highlight a positive regulatory role of *RhJAZ16* in salt stress adaptation, contrasting the conventional view of JAZ proteins as repressors in jasmonate signaling. Collectively, this study integrates genome-wide analysis with functional validation to advance our understanding of JAZ-mediated regulatory networks in rose, and provides a foundation for the genetic improvement of stress resilience in ornamental plants.

CRediT authorship contribution statement

Shikai Fan: Investigation, Formal analysis. **Fei Dong:** Resources, Investigation. **Jiao Zhu:** Writing – original draft, Investigation, Data curation, Conceptualization. **Rongchong Li:** Formal analysis, Data curation. **Yiyang Liu:** Writing – original draft, Supervision, Formal analysis, Conceptualization. **Chengpeng Wang:** Supervision, Funding acquisition.

Consent to publish

Not applicable.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.indcrop.2025.121872.

Data availability

Data will be made available on request.

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